SQL for Data Analytics

Total Time: 3 Hours

Theme: Retail Sales Exploration

Dataset: ***https://www.kaggle.com/datasets/vivek468/superstore-dataset-final***

Section A: Concept Application

1. What is the functional difference between SELECT * and specifying column names, and when is each preferred?
2. Which keyword renames a column in the output, and does this alias change the actual table structure in the database?
3. Why does wrapping a numeric value in quotes (e.g., '5000') in a WHERE clause create a data type conflict in SQL?
4. Contrast the results of ORDER BY Profit DESC versus ASC when the goal is to identify the top 10 most profitable orders.
5. What is the T-SQL equivalent of the LIMIT clause in MS SQL Server, and why does syntax vary across SQL engines?
6. Explain the logical execution order of a query containing SELECT, WHERE, ORDER BY, and LIMIT clauses.

Section B: Practical Task

1. Execute a query to retrieve the first 20 records from the orders table to verify data ingestion.
2. Select Order ID, Order Date, Sales, and Profit, applying a column alias to display Sales as Total_Sales.
3. Filter the dataset to isolate all high-value transactions where the Sales figure exceeds 5000.
4. Generate a report of the top 10 most profitable orders by sorting the records by Profit in descending order.

Section C: Mini Project

1. **Title:** Retail Profitability & Market Segment Analysis

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2. **Problem Statement:** Identify underperforming product categories and regions by analyzing the relationship between discount rates and net profit margins.
3. **Dataset Recommendation:** Sample Superstore Dataset (SampleSuperstore.csv) - https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
4. **Required Deliverables:** SQL script for database schema creation, multi-condition filtering queries, aggregated performance report by region, and a summary of loss-making transactions.

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